What Does Random Weight Search Buy? Votes, Prompts and First-Order Tilts in RandOpt

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Abstract

Random-perturbation post-training (RandOpt) samples thousands of Gaussian weight perturbations of a pretrained model, keeps the best on a small selection set and majority-votes them; we ask what this weight search buys. Running RandOpt at its own settings in four rows (GSM8K with two Qwen models and OLMo-2-1B; GQA with Qwen2.5-VL-3B), we find its gain is a vote, which sampling the unperturbed model supplies, plus a shift in answers that repairs damage done by RandOpt's own prompt (11.3 and 31.6 points in the two rows where it beats self-consistency). Choosing a prompt on RandOpt's own 200 selection questions (600 generations versus its 1,000,000) and then sampling matches or beats RandOpt in all four rows when a boxed-answer prompt is among the candidates; with public evaluation templates alone it does so in three of four, and RandOpt stays ahead on Qwen2.5-1.5B. With the chosen prompt held fixed, RandOpt's search adds nothing over sampling in any row, and what it selects is specific to the prompt it ran under. A first-order account with no fitted coefficients tracks how perturbations shift answer preferences ($r = 0.787$–0.938 across five models in two families), localizes the effect to middle language layers and explains why selection favours shared shifts, while failing for vision weights, large σ and chain-of-thought correctness; all confirmatory tests were specified in plan locks committed before their runs.

1. Introduction

Random-perturbation post-training (Gan & Isola, 2026) is a striking result. RandOpt samples thousands of Gaussian perturbations of a pretrained model's weights, keeps those that score best on a small selection set, and majority-votes their answers; it reportedly rivals PPO and GRPO on reasoning and vision benchmarks. Its authors read this as evidence that diverse task experts are dense around pretrained weights, a "neural thicket". The method reproduces, and the original paper itself notes that part of its GSM8K gain is a change in answer format. What is not yet known is what the weight search contributes beyond things that are much cheaper to obtain.

We ask two questions. *What does the weight search buy?* And *what do the selected perturbations change in the model's answers?* We answer the first with same-run comparisons at RandOpt's own settings, in four rows spanning two tasks and two model families, against self-consistency at the same test-time budget, plus controls that each isolate one component. We answer the second with a first-order account of how a perturbation shifts answer preferences, which we test directly and whose limits we map.

The answer is short (Figures 1 and 2). RandOpt's gain has two parts: a vote, and a shift in the model's answers shared across questions. Where the selected models barely beat the base model, sampling the unperturbed model votes better, and self-consistency beats RandOpt. Where RandOpt beats self-consistency, its own prompt damages the base model (by 11.3 and 31.6 points) and the selected perturbations recover part of that accuracy. Choosing among a few prompts on RandOpt's own 200 selection questions, at 0.06% of its selection compute, then sampling, matches or beats RandOpt in all four rows when the candidates include a format-fixing prompt, and in three of four with public evaluation templates alone. With the chosen prompt held fixed, RandOpt's search adds nothing over sampling in any row, and its selected models are no better than the base model on test. What selection finds is specific to the prompt it runs under.

Contributions.

1. Same-run comparisons of RandOpt with self-consistency at RandOpt's settings in four rows (two model families, two tasks), reproducing its published numbers and replicating both of its wins with a second population (Section 3.1).

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2. A decomposition of the gain into a vote and a shared shift, and evidence that the shift repairs damage done by RandOpt's own prompt (Sections 3.2–3.3).

3. A practical baseline and its condition: prompt selection on the same selection data, then sampling, matches or beats RandOpt when a format-fixing candidate is included; with the prompt held fixed, the search adds nothing in all four rows (Sections 3.4–3.5).

4. A first-order account of perturbation-induced answer shifts, validated across five models in two families, with demonstrated limits, and an explanation of why selection favours shared shifts (Sections 4–5).

Confirmatory tests were specified in plan locks committed before their corresponding runs; exploratory analyses are identified separately. Every lock, all per-item outputs and the analysis code are released with the paper.

2. Background and Setup

RandOpt. Given a pretrained model with weights θ_0, RandOpt (Gan & Isola, 2026) draws N seeds and noise scales $\sigma \in \{0.0005, 0.001, 0.002\}$, forms $\theta_0 + \sigma\epsilon$ with ϵ a Gaussian vector generated from the seed, scores each perturbed model by its reward on a selection set of 200 training questions under greedy decoding, and answers each test question by a majority vote over the greedy answers of the top K. We use its settings throughout: N=5000, K=50 (K=10 as a secondary), its prompts, answer extractors, scorers and vote rule. This costs $5000 \times 200 = 1{,}000{,}000$ greedy generations for selection and 50 per test question.

Rows. GSM8K (Cobbe et al., 2021) with Qwen2.5-1.5B-Instruct and Qwen2.5-3B-Instruct (Qwen Team, 2024) (the original paper's models) and with OLMo-2-0425-1B-Instruct (Team OLMo, 2024) (a second family), evaluated on all 1319 test questions; and GQA (Hudson & Manning, 2019) with Qwen2.5-VL-3B-Instruct (Qwen Team, 2025), evaluated on 1238 testdev questions whose images are disjoint from the selection questions' images. The GSM8K rows run RandOpt's released script; its recomputed vote equals the count it prints. The released code cannot pass images to the model, so the GQA rows use a re-implementation built on RandOpt's own perturbation, scoring and voting components. All model revisions are pinned (Appendix B).

Baseline. Self-consistency (Wang et al., 2023) (SC@K) majority-votes K samples of the *unperturbed* model at temperature 0.7, with RandOpt's scorer and vote rule. SC@50 matches RandOpt's test-time budget of 50 generations per question; RandOpt's selection generations are not charged to it.

Statistics and protocol. Every comparison is paired on the same test questions. We report differences D in accuracy points with 95% confidence intervals from a paired item bootstrap (10,000 resamples); a method is *ahead* when the interval excludes zero, and two methods are *equivalent* when it lies within ± 2 points. Confirmatory tests were specified in plan locks committed before their corresponding runs; exploratory analyses are identified separately. Each lock fixes the comparison, its decision rule and its gates: a base-model reproduction or environment check against an earlier independent measurement, a fidelity check against RandOpt's own logs where a re-implementation is used, and a budget rule that stops a run rather than shrink it. Appendix A lists every locked test with its outcome, including failed gates and runs that were not completed.

3. What Weight Search Buys

We run RandOpt at its own settings in four rows and compare it with self-consistency on the same test questions at matched test-time budget (Table 1). We then ask what RandOpt's gain consists of, and whether something cheaper than weight search supplies it.

3.1. Same-run comparison under RandOpt's own prompts

RandOpt reproduces: on GSM8K its K=50 vote scores 77.18 with Qwen2.5-1.5B and 86.66 with Qwen2.5-3B, against the published 76.4 and 87.1. Under RandOpt's own prompts, the comparison with SC@50 splits two ways. In the two Qwen GSM8K rows self-consistency is ahead ($D = -2.65$ [-4.32, -0.99] and -1.59 [-2.65, -0.53]). On GQA RandOpt is ahead ($+3.47$ [$+1.62$, $+5.41$]), and a second search with a different population seed replicates this ($+3.63$ [$+1.78$, $+5.49$]; seed-to-seed difference $+0.16$ [-0.97, $+1.29$]). On OLMo-2-1B, a model outside the original study, RandOpt is ahead by $+8.72$ [$+6.75$, $+10.77$], and a second search with a different population, run with our re-implementation, replicates this ($+10.39$ [$+8.26$, $+12.51$]). The published GQA number (69.0) uses all of testdev with train-split selection and is not comparable to our image-disjoint split.

Under its own prompts, then, RandOpt is a real improvement over sampling in two of four rows. The rest of this section asks what that improvement is.

3.2. A vote and a shared shift

RandOpt's gain can come from two places: each selected model may be better than the base model, or the vote over them may correct errors that single models make. The mean single-model accuracy of the 50 selected models separates the two. In the Qwen rows, where RandOpt loses, the selected models are only $+4.0$ (64.3 vs 60.3) and $+0.2$

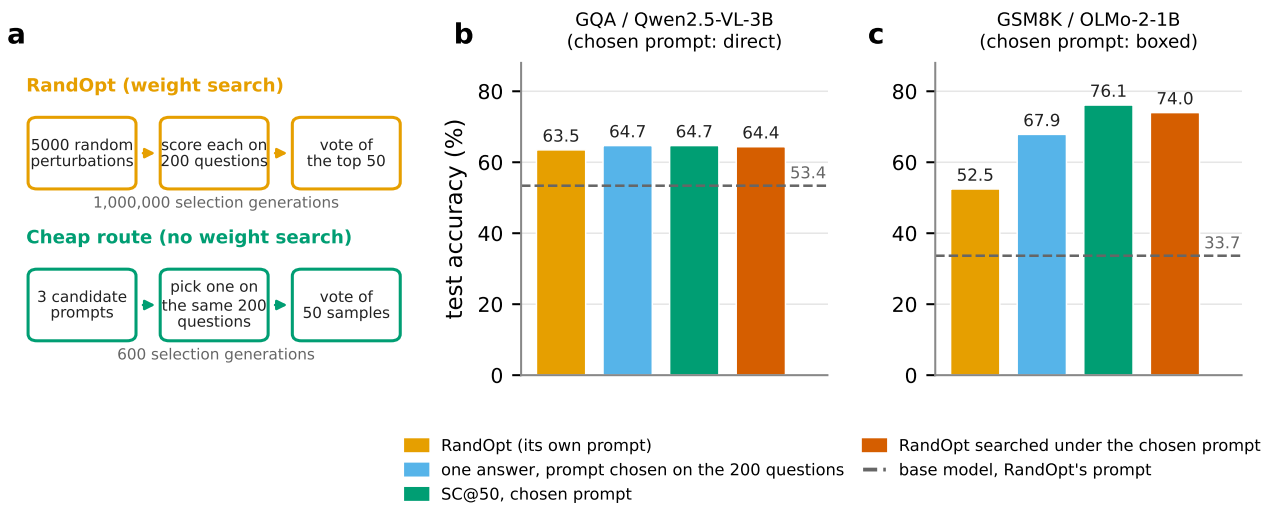


Figure 1. **Choosing a prompt on RandOpt's own selection questions matches or beats its weight search, at 0.06% of the selection compute.** (a) RandOpt scores 5000 random weight perturbations on 200 selection questions and votes the top 50; the cheap route picks one of three prompts on the same questions and votes 50 samples of the unperturbed model. (b, c) The two same-run rows where RandOpt beats self-consistency under its own prompt. RandOpt's prompt damages the base model (dashed); one answer under the chosen prompt already matches (GQA) or beats (OLMo) RandOpt, and re-running RandOpt's search under the chosen prompt adds nothing over sampling. GQA uses a faithful re-implementation of RandOpt.

(80.9 vs 80.7) points above the base model individually: the gain is almost all vote, and sampling the unperturbed model votes better. On GQA and OLMo, where RandOpt wins, the selected models are about 5 points above base individually (58.4 vs 53.4; 40.3 vs 35.25). That is a shift in answers shared across questions, which sampling cannot supply, since every sample comes from the unchanged model.

Two exploratory measurements qualify this. Measured as the vote's gain over its own members' mean accuracy, sampling's vote adds more than RandOpt's in three rows (Qwen-1.5B $+21.7$ vs $+12.9$; Qwen-3B $+9.9$ vs $+5.8$; GQA $+10.0$ vs $+5.1$) and less on OLMo ($+10.1$ vs $+12.1$). And perturbations that beat the base model on the 200 selection questions are common, 11.8–58.3% of the 5000 depending on the row; the population mean is level with base on the Qwen rows (-0.4 and -0.3 points) and below it on GQA and OLMo.[1]

3.3. The shift repairs damage done by RandOpt's own prompt

GQA. RandOpt's GQA prompt asks for step-by-step reasoning. Asking the base model for a direct answer instead raises its greedy accuracy from 53.39 to 64.70 ($+11.31$ [$+8.48$, $+14.14$]). The selected perturbations largely stop reasoning (exploratory): their median generation is 10 tokens against the base model's 146, 43.7% of their outputs contain step-by-step text against 97.0%, and across the 50 members accuracy and mean length correlate at $r = -0.89$. On the 616 test questions where the two base prompts disagree, the members give the direct prompt's answer 48.7% of the time and the step-by-step prompt's 15.3%. Truncation explains only part of the gain: raising the token limit from 256 to 1024 shrinks RandOpt's advantage by 0.97 [0.32, 1.70] points but leaves it at $+2.67$ [$+0.73$, $+4.52$]. A single direct-prompt generation of the base model scores 64.70, against RandOpt's 63.49 (-1.21 [-2.83, $+0.40$], no difference detected).

OLMo. RandOpt's GSM8K instruction ("...output the final answer after ####") halves OLMo-2-1B's accuracy: greedy 33.66, against 65.28 with the question alone and 67.85 with a boxed-answer instruction. Under the plain prompt, SC@50 is ahead of RandOpt by 22.21 points (-22.21 [-24.87, -19.56]), and a single plain generation by 12.81 [9.86, 15.69]. Unlike on GQA, the selected members do not answer like the better-prompted model (exploratory): on the 878 questions where the boxed and RandOpt-prompt base answers disagree, they side with each about equally often (24.1% vs 27.7%). Here the shift recovers accuracy without reproducing the better prompt's answers.

Damage accompanies RandOpt's wins but does not decide them. Our locked prediction was that RandOpt beats

[1] For the Qwen rows we use the base selection accuracy measured by the same engine that reproduces RandOpt's perturbed rewards (68.0 and 84.0). RandOpt's script prints 73.00 and 85.50 for the same 200 prompts, reproducibly across engine counts, while its test-set base agrees with ours (60.27 vs 60.05); we treat the print as engine-specific (Appendix C). No confirmatory comparison uses it.

SC@50 where its prompt damages the base model relative to a plain or direct prompt. It held in all four rows: the damage was +11.3 (GQA) and +31.6 (OLMo) points where RandOpt won, and +4.70 [+1.97, +7.43] and −7.20 [−9.55, −4.78] for Qwen-1.5B and Qwen-3B, where it lost. For the Qwen models the plain prompt is no fix: SC@50 under it trails RandOpt (+2.58 [+0.53, +4.62] and +7.05 [+5.23, +8.95]). Measured post hoc against the prompt chosen in Section 3.4, Qwen-1.5B's damage is +10.2 and RandOpt still lost (Appendix C, Figure 6), so damage alone does not decide the outcome.

3.4. Choosing a prompt on RandOpt's own selection data

The controls above fix a prompt after seeing results. A practitioner can instead choose the prompt the way RandOpt chooses perturbations: on its 200 selection questions. We fixed three candidates per task (GSM8K: RandOpt's prompt, the question alone, a boxed-answer instruction; GQA: RandOpt's step-by-step prompt, a direct-answer prompt, a short-answer prompt; Appendix H). Per row, we chose the candidate with the highest greedy base accuracy on the selection questions and ran SC@50 under it on test. This costs 600 greedy generations against RandOpt's 1,000,000 (0.06%). The rule chose the boxed prompt in all three GSM8K rows and the direct prompt on GQA.

Prompt selection followed by sampling matches or beats RandOpt in all four rows (Figure 2, Table 1). It is ahead on Qwen-1.5B (−2.96 [−4.62, −1.29]) and OLMo (−23.65 [−26.23, −21.08]). No difference is detected on Qwen-3B (−0.38 [−1.67, +0.91], equivalent within ±2 points) or on GQA (−1.21 [−2.83, +0.40], not equivalent). The ordering holds under strict answer scoring and McNemar tests with Holm correction, and with a K=10 vote, where RandOpt is ahead in no row (exploratory; Appendix C). It does not depend on the sampling temperature: at $T = 0.5$ and 1.0 the two no-difference rows stay no-difference (pre-registered; Appendix C).

Table 1 also separates what the prompt does from what the vote does. One greedy answer under the chosen prompt already beats RandOpt on OLMo (67.9 vs 52.5) and matches it on GQA (64.7 vs 63.5), so there the prompt does the work; on GQA voting then adds nothing (SC@50 64.7). On the Qwen rows the single answer is below RandOpt (70.2 vs 77.2; 82.3 vs 86.7) and voting closes the gap (80.1; 87.0). The choice itself is stable at RandOpt's selection-set size (exploratory): resampling the 200 selection questions, the rule picks the same prompt in 99.6%, 95.5%, 77.5% and 99.6% of bootstrap draws, and where it switches on OLMo the alternative is nearly as good. With 50 questions the Qwen-3B choice becomes fragile (73%), and its runner-up, the plain prompt, loses to RandOpt by 7.05 points (Appendix C).

Does the result depend on our candidates? The three candidates were fixed before this test but chosen after earlier results. We therefore repeated it with candidates we could not have tuned: public evaluation templates from lm-evaluation-harness, simple-evals, LLaVA and BLIP-2, fixed in a plan lock before any of them was run, and chosen by the same rule (Figure 2, diamonds). Prompt-selected sampling then matches or beats RandOpt in three rows: Qwen-3B (−0.45 [−1.67, +0.76], equivalent), OLMo (−5.23 [−7.66, −2.81]) and GQA (−2.67 [−4.60, −0.73], now ahead). On Qwen-1.5B no public template beats RandOpt's own prompt on the selection questions (67.5 vs 68.5), and RandOpt is ahead (+4.17 [+1.97, +6.44]). Pooling all six candidates, the rule picks the boxed prompt for the GSM8K rows and the LLaVA template for GQA, and matches or beats RandOpt in all four rows. The cheap route therefore needs a candidate that fixes the answer format; a boxed-answer instruction, standard for math, is one.

3.5. With the prompt held fixed, the search adds nothing

Prompt choice and weight search can be combined: RandOpt can be run under the chosen prompt. We re-ran its search, with the same 5000 perturbations as in Section 3.1, under the direct prompt on GQA and the boxed prompt in the three GSM8K rows (the latter with a fast re-implementation that passed its environment and fidelity gates). In every row the top 50 again beat the base model on the selection questions (for example GQA 68.0–69.5% vs 66.0%; OLMo 84.0–86.0% vs 79.5%). On test, though, the selected models were no better than base in any row (−0.60 [−1.27, +0.05] on GQA, −0.81 [−2.19, +0.57] on OLMo, −0.47 [−1.45, +0.53] on Qwen-3B, −0.51 [−1.71, +0.73] on Qwen-1.5B), and RandOpt's vote never beat sampling (Table 1, Figure 2, triangles). It was equivalent on GQA (−0.32 [−1.29, +0.65], within ±2 points) and behind on OLMo (−2.12 [−3.49, −0.83]), Qwen-3B (−1.36 [−2.35, −0.38]) and Qwen-1.5B (−4.32 [−5.91, −2.81]). On OLMo the better prompt lifts RandOpt itself by 21.53 points [18.88, 24.18], yet it still trails sampling. The top 50 found under the two prompts share no member in three rows and three members on Qwen-1.5B (Section 5).

With the prompt held fixed, then, weight search added nothing measurable over sampling the unperturbed model in any of the four rows. (The Qwen-1.5B search first failed a gate that compared our engine with the base accuracy printed by RandOpt's script; we traced that print to an engine-specific effect, re-ran the search under gates that compare like with like, and report the failed attempt in the ledger; Appendix A.)

Table 1. **Same-run results** (N=5000, K=50; accuracy in %, differences in points with paired 95% CIs). SC: self-consistency over 50 samples of the unperturbed model; 1 gen.: one greedy answer of the unperturbed model (same engine as SC), so within a prompt block SC@50 − 1 gen. is the voting effect and across blocks 1 gen. is the prompt effect. The chosen prompt (in parentheses) maximizes base accuracy on RandOpt's 200 selection questions among our three candidates; the public template is chosen by the same rule among public evaluation templates fixed before any was run (Section 3.4).

	RandOpt's prompt				Prompt chosen on selection set			Public template chosen		Search under chosen prompt	
Row	1 gen.	SC@50	RandOpt	RandOpt − SC	1 gen.	SC@50	RandOpt − SC	SC@50	RandOpt − SC	RandOpt	RandOpt − SC
GSM8K / Qwen2.5-1.5B (boxed)	60.0	79.8	77.2	−2.65 [−4.32, −0.99]	70.2	80.1	−2.96 [−4.62, −1.29]	73.0	+4.17 [+1.97, +6.44]	75.8	−4.32 [−5.91, −2.81]
GSM8K / Qwen2.5-3B (boxed)	80.1	88.2	86.7	−1.59 [−2.65, −0.53]	82.3	87.0	−0.38 [−1.67, +0.91]	87.1	−0.45 [−1.67, +0.76]	85.7	−1.36 [−2.35, −0.38]
GQA / Qwen2.5-VL-3B (direct)	53.4	60.0	63.5	+3.47 [+1.62, +5.41]	64.7	64.7	−1.21 [−2.83, +0.40]	66.2	−2.67 [−4.60, −0.73]	64.4	−0.32 [−1.29, +0.65]
GSM8K / OLMo-2-1B (boxed)	33.7	43.7	52.5	+8.72 [+6.75, +10.77]	67.9	76.1	−23.65 [−26.23, −21.08]	57.7	−5.23 [−7.66, −2.81]	74.0	−2.12 [−3.49, −0.83]

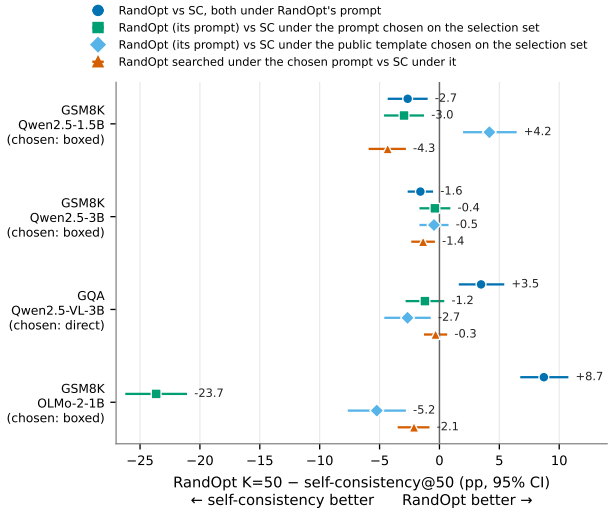


Figure 2. **Choosing a prompt on RandOpt's own selection data matches or beats its weight search in every row, and with that prompt held fixed the search adds nothing.** RandOpt (K=50) minus self-consistency (K=50) per row, paired 95% CI: both under RandOpt's prompt (circles); SC under the prompt chosen on the 200 selection questions (squares) or under the public template chosen the same way (diamonds; on Qwen-1.5B no public template beats RandOpt's prompt and RandOpt is ahead); and RandOpt's search itself run under the chosen prompt (triangles). GQA uses a faithful re-implementation of RandOpt.

4. What a Perturbation Does to Answers

Section 3 treated the selected perturbations as black boxes. Here we ask what a single perturbation does to a model's answers, in the regime where that can be measured cleanly: direct answers, where each question has a well-defined preference between candidate answers.

A first-order prediction. Let $c_j(\theta)$ be a contrast for question j, for example the log-probability of one answer option minus another, or a tilt averaged over questions toward options containing a given content word. A perturbation $\theta_0 + \sigma\epsilon$ changes it by

$$\Delta c_j \approx \sigma \langle g_j, \epsilon \rangle, \qquad g_j = \mathrm{fold}\big(\nabla_\theta c_j(\theta_0)\big), \qquad (1)$$

where fold maps each parameter tensor's gradient onto the coordinates of RandOpt's shared noise stream (Appendix D). The prediction needs the base model's gradient and the perturbation's own noise; it has no fitted coefficients. We reconstructed every perturbation byte-exactly from RandOpt's generator and verified it against the engine's weights.

It tracks the measurements across models and tasks (Figure 3). For content tilts on two OmniSpatial tasks, prediction and measurement correlate at $r = 0.938$ and 0.915 (Qwen3-VL-8B), and at 0.925 for Qwen2.5-VL-7B, after a first attempt failed its reliability gate (0.52) and a remedy fixed in advance was applied (slopes 0.83–0.92 across these three tests). Per question, in RandOpt's own GQA setting with direct answers, $r = 0.930$ for $\sigma \leq 0.002$. On a different family and a text-only task, OLMo-2-1B and OLMo-2-7B on ARC-Challenge (Clark et al., 2018), $r = 0.790$ and 0.787 (slopes 0.76 and 0.77). The law also holds item by item rather than only on average: the pooled per-item correlation is 0.771, with a per-item median of 0.88. In the first test, 99% of the measured tilt variance is answer content, not letter position. The prediction is first-order, not exact: slopes are below one and the scatter is real.

It is localized to the middle language layers (Appendix E, Figure 7). Inserting the perturbation one block at a time, the block-level prediction holds at $r = 0.978$ (slope 0.95), contributions add up ($r = 0.96$), and layers 6–23 of 36 carry 0.91 of the measured tilt variance (0.41, 0.22 and 0.28 for layers 6–11, 12–17 and 18–23; predicted 0.44, 0.22, 0.23). The middle layers carry most of the predicted variance for a second task (0.73) and a second model (0.77).

Its limits are as clear as its successes. (i) *Vision weights*: with the vision encoder perturbed too, the whole-model first-order prediction of per-example contrasts fails its locked test ($r = 0.137$, falsified); the vision group alone gives $r = 0.05$, while language groups give 0.71–0.96. (ii) *Noise scale*: at $\sigma = 0.005$, beyond RandOpt's grid, the law breaks in every model we tested ($r = 0.21$–0.31), consistent with second-order drift. (iii) *Chain of thought*: under RandOpt's step-by-step GQA prompt, 22% of question–perturbation pairs flip correctness at $\sigma \leq 0.002$, unrelated to the first-order contrast (sign agreement 0.52). Perturbations and sampling flip similarly susceptible questions (Spearman 0.946 between per-question flip propensities; Appendix E), but perturbed models carry persistent effects across question

sets ($r = 0.74$), so perturbation is not pure re-sampling. The accounts of Sections 4 and 5 therefore cover the direct-answer regime; Section 3 rests on its own experiments.

5. Why Selection Finds Shared Shifts

Selection gains transfer only where the prompt leaves accuracy unclaimed (exploratory; Figure 4). In every one of our ten N=5000 searches, the top 50 beat the base model on the 200 selection questions, while the population mean is at or below base. Whether that gain survives on test depends on the prompt. Under a prompt that damages the base model by 10 points or more, the selected models are better on test by 4.0–7.9 points (GQA and OLMo, two searches each, and Qwen-1.5B); under RandOpt's prompt for Qwen-3B, which costs little, by 0.2; and under every chosen prompt they are slightly worse than base (-0.5 to -0.8). In every search, selection found models that look better on 200 questions; only a shared, prompt-repairing shift carried over.

What selection finds is specific to the prompt (exploratory; Figure 5). Scoring the same 5000 perturbations under two prompts, their selection rewards are nearly unrelated (Spearman 0.025 on OLMo, 0.171 on GQA), and the winners under RandOpt's prompt fall to the 37th and 31st percentile under the better prompt. On GQA the selected perturbations reproduce the direct prompt's answers: where the two base prompts disagree, they side with the direct one 48.7% of the time against 15.3%. On OLMo they do not (24.1% vs 27.7%). In our settings, then, the dense "experts" near the pretrained weights are repairs for a particular prompt rather than task experts.

Why selection favours shared shifts. Under Equation 1, a perturbation moves question j's margin by a Gaussian amount with standard deviation $\sigma\|g_j\|$ (Appendix D). Three consequences follow, which we checked against criteria fixed before computing them (exploratory, on existing data). A question flips with probability $\Phi(-|c_j|/\sigma\|g_j\|)$, which predicts observed flips (AUC 0.935, calibrated within a factor of 1.41). An unselected vote returns the base answer, as it does on 96 of 96 questions. And top-K selection acts as a noisy first-order step along the selection set's folded gradient: it rewards directions shared across the selection questions, such as a format or answer-preference shift, and otherwise fits selection-set noise. This last step is derived, not tested directly. The account covers direct answers at small σ; that the shared shifts we found are ones a prompt also gives is an empirical finding of Section 3, not a consequence of the theory.

A case study at its boundary. One OmniSpatial search (Qwen3-VL-8B, N=5000) produced a winner whose $+8.0$ point gain on its re-ranking set sat entirely on a question format (compass directions) that the official test set nearly lacks (2 of 561 items); on fresh items matched to the selection data it keeps $+2.67$ [$+0.17$, $+4.93$] (Appendix F). Selection had favoured a tilt toward the answer content the selection labels reward, as the theory predicts. But removing that content shift leaves $+2.33$ of the $+2.67$ (not explained), and the full first-order prediction does not explain which items the winner changes ($r = 0.16$, failing its locked test). The winner's gain is also prompt-specific: under the benchmark's step-by-step prompts it becomes -0.67 [-3.95, $+2.51$] or -3.83 [-6.81, -0.84], and a prompt chosen on the winner's own selection data gives the unperturbed model an accuracy at least as high (-1.50 [-5.35, $+2.23$], no difference detected, not equivalent).

6. Related Work

Random weight search. RandOpt (Gan & Isola, 2026) belongs to the family of zeroth-order, population-based methods such as evolution strategies (Salimans et al., 2017), applied here as a single round of sampling and selection around a pretrained model. Its authors already separate format from reasoning gains on GSM8K ("format thickets"); our results agree with that reading and quantify it against cheap baselines. Concurrent work finds that Bayesian optimization within a random linear embedding of weight space matches or exceeds RandOpt with five times fewer candidate evaluations (Cendra et al., 2026). Informal analyses argue that the thicket reflects unused parameter capacity rather than pre-solved tasks (Atanassov, 2026), and that RandOpt's advantage over gradient methods grows with output sequence length (Liu, 2026). We add same-run baselines, controls that isolate each component, and a tested account of what single perturbations do.

Voting and prompts. Self-consistency (Wang et al., 2023) is the natural baseline for any method that votes over generations at test time; we find it supplies most of what RandOpt's vote supplies. Language models are sensitive to prompt format (Sclar et al., 2024); RandOpt's wins coincide with prompts that damage the base model, and choosing among a few prompts on the selection data recovers them. Weight averaging (Wortsman et al., 2022) combines models in weight space rather than voting over their outputs.

Linear response. Our first-order account is a linearization of the model around its pretrained weights in the direction of each perturbation. Its use here is narrow: it predicts answer-preference shifts with no fitted coefficients, and we report where it fails.

7. Discussion, Recommendations and Limitations

What weight search buys, in our settings. A vote, and shifts in answer form shared across questions. The vote

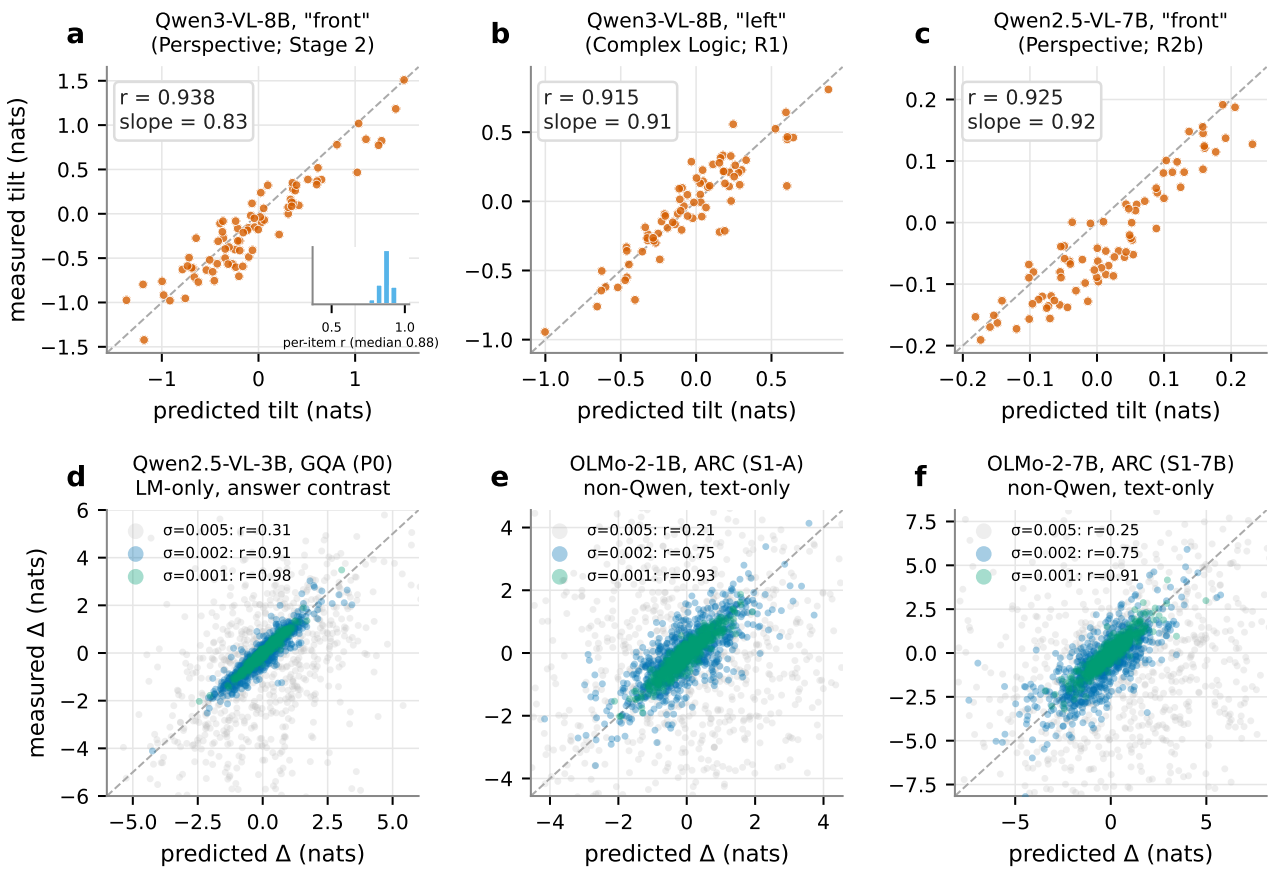


Figure 3. **A first-order prediction with no fitted coefficients tracks how perturbations shift answer preferences, across five models in two families.** (a–c) Answer-content tilts on two OmniSpatial tasks and two Qwen-VL models (slopes 0.83–0.92). (d–f) Per-question answer contrasts in RandOpt's GQA setting and on ARC-Challenge with the non-Qwen OLMo-2-1B and 7B, coloured by noise scale: the account holds at $\sigma \leq 0.002$ and breaks at σ=0.005. Inset: per-item correlations (median 0.88).

is available from sampling; the shifts were, in every row we tested, available from a prompt chosen on the same 200 selection questions, provided a format-fixing prompt is among the candidates. With that prompt held fixed, the search added nothing in any row, and what selection found was specific to the prompt it ran under. This does not show that nearby better models are absent: perturbations that beat the base model on the selection set are common (11.8–58.3%), and under a damaging prompt the selected ones are better on test. It shows that, at RandOpt's settings, what selection recovers is what a prompt also recovers.

Recommendations for evaluating weight-space search. (1) Compare against self-consistency in the same run, at the same test-time budget, with the same scorer. (2) Include a prompt chosen on the selection data as a baseline, with a format-fixing candidate. (3) Report the selected models' individual accuracy next to the vote's, so the vote and the shift can be told apart. (4) Report selection compute. (5) Check that the base model reproduces before comparing, and compare like with like: in one case the same 200 prompts gave a base accuracy of 73.0% in RandOpt's script and 68.0% in our engine, reproducibly.

Limitations. Two tasks and models up to 8B parameters; one RandOpt search per Qwen GSM8K row under its own prompt (two on GQA and on OLMo), each row with a further search under the chosen prompt; RandOpt's selection set of 200 questions throughout, so we do not test whether larger selection sets change the picture; our prompt candidates were chosen after earlier results, which the public-template test addresses for three of four rows; the first-order account covers direct answers at small σ, not chain-of-thought correctness or vision weights, and does not explain most of the OmniSpatial winner's gain. Exploratory analyses are labelled as such throughout.

Impact Statement

This paper re-examines an evaluation and proposes cheaper baselines for it. It introduces no new capability. Its main practical consequence is that a prompt chosen on a small

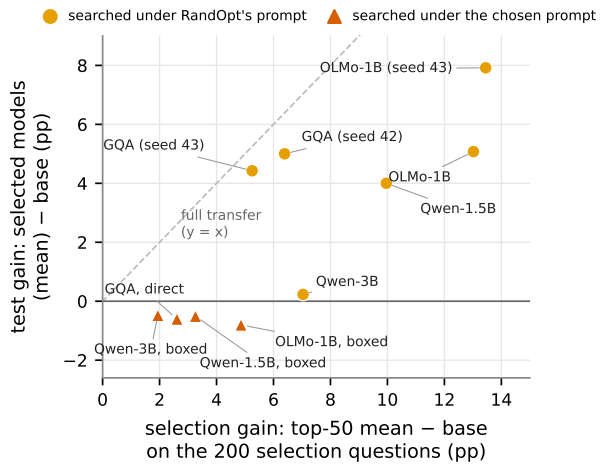


Figure 4. **Selection gains transfer only where the prompt leaves accuracy unclaimed** (exploratory). Every search's top 50 beats the base model on the 200 selection questions; the selected models are better on test only under a damaging prompt, and in none of the four searches run under the chosen prompt (triangles).

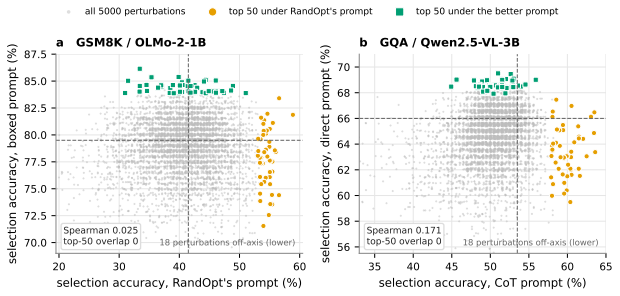


Figure 5. **The selected "experts" are specific to the prompt** (exploratory). The same 5000 perturbations scored under two prompts: rank correlations 0.025 (OLMo) and 0.171 (GQA); the two top-50 sets share no member, and RandOpt's winners fall below the base model (dashed) once the prompt is improved.

selection set, followed by sampling, can replace a weight search that costs more than 1,600 times as much selection compute, which reduces the compute spent on post-training experiments of this kind.

References

Atanassov, T. A thicket by any other name. https://noxidog.substack.com/p/a-thicket-by-any-other-name, March 2026. Blog post.

Cendra, N. B., Ezzerg, A., Cendra, F. J., Knoblauch, J., and Zeitler, J. Can Bayesian optimization efficiently find a strong single expert in neural thickets? *arXiv preprint arXiv:2608.10867*, 2026.

Clark, P., Cowhey, I., Etzioni, O., Khot, T., Sabharwal, A., Schoenick, C., and Tafjord, O. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Gan, Y. and Isola, P. Neural thickets: Diverse task experts are dense around pretrained weights. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, 2026. arXiv:2603.12228.

Hudson, D. A. and Manning, C. D. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6700–6709, 2019.

Jia, M., Qi, Z., Zhang, S., Zhang, W., Yu, X., He, J., Wang, H., and Yi, L. OmniSpatial: Towards comprehensive spatial reasoning benchmark for vision language models. In *International Conference on Learning Representations*, 2026. arXiv:2506.03135.

Liu, Z. When does RandOpt work? https://kindxiaoming.github.io/blog/2026/randopt/, March 2026. Blog post.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

Qwen Team. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025.

Salimans, T., Ho, J., Chen, X., Sidor, S., and Sutskever, I. Evolution strategies as a scalable alternative to reinforcement learning. *arXiv preprint arXiv:1703.03864*, 2017.

Sclar, M., Choi, Y., Tsvetkov, Y., and Suhr, A. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations*, 2024.

Team OLMo. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2024.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.

Wortsman, M., Ilharco, G., Gadre, S. Y., Roelofs, R., Gontijo-Lopes, R., Morcos, A. S., Namkoong, H., Farhadi, A., Carmon, Y., Kornblith, S., and Schmidt, L. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 23965–23998, 2022.

A. Confirmatory-Test Ledger

Every locked test, with the rule fixed in its lock and its outcome, including falsified, inconclusive, gated-out and skipped tests. A lock is the commit that fixed the rule before the output existed; amendments are recorded in the same folders. Studies: GPU-A, Stage 1–3, R1, R2/R2b, I5, M1, C1, τ-check, P0–P3, S1, S1-7B (the first-order account and the OmniSpatial case study); C, C3B, G2, G2R, O1, PS, OS, GD, GB, Q2, O2, G4, G3, RV (Section 3). Engineering failures that produced no data (an offline-mode bug, a package conflict, a tar-ownership error, a budget rule that skipped a RandOpt arm on a one-GPU pod) and the four pod interruptions during the RV runs are recorded in the repository alongside the locks.

Study	Lock	Test	Statistic	Rule (fixed in the lock)	Result	Outcome
GPU-A run 1	198c352	S2a HF vs vLLM fidelity	median per-item max \—Δlog p\—; argmax agreement	≤ 0.10; $\geq 99\%$	0.25; 98%	**NO-GO** (BF16 output resolution; led to amendment A1)
GPU-A A1	79252d4	V0 score-definition gate; S2b candidate bytes	fp32 contrast checks; packed-tensor SHA	as locked	pass; 642/642	pass
GPU-A A1	79252d4	F1 whole-model first-order (vision perturbed)	pooled r, 13 candidates $\times$ 761 items	≥ 0.6 support; < 0.3 falsify	0.137	**falsified**
GPU-A A1	79252d4	F2 changed answers predicted to flip	share	< 0.5 falsify	0.53	not falsified (weak)
GPU-A A1	79252d4	F3 RERANK gain prediction	r over 543	≥ 0.5 support; < 0.3 falsify	0.376	inconclusive
GPU-A A1	79252d4	S2d group partials vs insertions	pooled r	< 0.5 falsify	0.239	**falsified**
Stage 1+2	09c1f1f	S1a reliability; S1b fidelity	Spearman–Brown; exactness	≥ 0.7; all exact	0.878; exact	pass
Stage 1+2	09c1f1f	**S2 tilt law** (Qwen3-VL-8B)	r(pred_NOV, T_NOV), 80 perturbations	≥ 0.6 pass; < 0.3 falsify	0.938 (slope 0.83)	**pass**
Stage 1+2	09c1f1f	S2 share carried by non-vision	r(T_NOV, T_FULL)^2	≥ 0.5	0.843	pass
Stage 3	ce60c5d	3A-0 pipeline consistency	r	≥ 0.9	0.938	pass
Stage 3	ce60c5d	3A-1 τ vs SEARCH gain	r over 5000	≥ 0.25 pass; < 0.10 falsify	0.240	inconclusive
Stage 3	ce60c5d	3A-2 τ vs RERANK gain	r over 543	same	0.402	pass
Stage 3	ce60c5d	3A-3 enrichment of advanced candidates	mean z difference	≥ 0.3 SD	+0.09 [−0.01, 0.18]	not supported
Stage 3	ce60c5d	3B-1 block-level law	pooled r, 160 pairs	≥ 0.6 pass; < 0.3 falsify	0.978 (slope 0.95)	**pass**
Stage 3	ce60c5d	3B-2 additivity; 3B-3 dominance	r; variance share	≥ 0.9; $\geq 50\%$ for one block	0.96; none	valid; distributed
R1	00155cc	reliability; position share; **law**	SB; share; r	≥ 0.7; < 0.20; ≥ 0.6	0.866; 0.184; **0.915**	**replicated**
R1	00155cc	middle-layer share	share of predicted variance, layers 6–23	≥ 0.5	0.73	supported
R2	47d2102	reliability; **law** (Qwen2.5-VL-7B, 96 items)	SB; r	≥ 0.7; ≥ 0.6	**0.518**; 0.920	**inconclusive** (reliability gate failed)
R2b	1cb3252	reliability; position share; **law** (363 items; remedy fixed in advance)	SB; share; r	≥ 0.7; < 0.20; ≥ 0.6	0.949; 0.073; **0.925**	**replicated**
I5	a1c2a7a	consistency gate; **per-item law**	r; pooled r, 7680 pairs	≥ 0.99; ≥ 0.6 item-level, < 0.3 average-only	0.99999; **0.771**	**item-level**
M1	5eb864b	M1-1 winner on fresh matched items	gain, image-cluster CI	transfers if CI > 0; full if also $\geq +6$	+2.67 [0.17, 4.93]	transfers (not full)
M1	5eb864b	M1-2 gold-front concentration	gain difference, CI	supports if CI > 0	+1.95 [−2.57, 6.47]	not supported
M1	5eb864b	M1-3 τ vs fresh gain	r over 59 (Fisher CI)	≥ 0.25 supports	0.61 [0.42, 0.75]	supports
M1	5eb864b	M1-4 top-50 vs controls	mean difference, bootstrap CI	reported	+1.16 [0.31, 2.03]	reported
C1	5aa4c06	**C1-1 prior calibration of the winner**	calibrated vs raw fresh gain	explains if $\leq 0.5\times$raw with CI; not explained if $\geq 0.8\times$raw	2.33 vs 2.67	**not explained**
C1	5aa4c06	C1-2 surviving share (candidates with raw g > 0)	Σg_cal / Σg	≤ 0.5 supports	≈ 0.00 [−0.42, 0.32]	locked label "supports"; **flagged**: a cross-fitted exploratory check shows it is regression-to-mean, so it is not evidence
τ-check	8c52ba1	criteria fixed before computing (exploratory data)	within-top-50 r; partial r; control sign	≥ 0.30; ≥ 0.20; ≥ 0	0.59; 0.31; 0.57	holds (works through the front tilt)
P0	d5152f7	**per-question first-order** (LM-only GQA)	pooled r, $\sigma \leq 0.002$	≥ 0.5 GO; < 0.3 NO-GO	**0.930**	**GO** (σ = 0.005: 0.31)
P1	464aed0	P1-A flip propensities	Spearman over 400 items	≥ 0.6 supports	0.946	supports
P1	464aed0	P1-B persistence	r(selection, held-out) over 64	persistent if ≥ 0.3 with CI > 0	0.74 [0.61, 0.84]	**persistent** (rejects pure resampling)
P1	464aed0	P1-C top-8 vote vs SC at T*	difference, item bootstrap CI	advantage if CI > 0	+2.5 [−3.5, 8.5]	no advantage (underpowered)
P2	01efd9d	base gate; SC@50 vs published RandOpt, GSM8K-1.5B	greedy diff; SC − RandOpt	± 2.0; match if ≥ -1.0	0.5; +3.4	gate pass; MATCHES (label)
P2	01efd9d	same, GSM8K-0.5B / GQA-VL-3B	greedy diff	± 2.0 / ± 2.5	+3.3 / −2.6	gate **fail**: not comparable
P3	c60b600	base gate; SC@50 vs published RandOpt, GSM8K-3B	greedy diff; SC − RandOpt	± 2.0; ≥ -1.0	0.5; +1.1	gate pass; MATCHES (label)

Study	Lock		Test	Statistic	Rule	Result	Outcome
P3	c60b600		same, MATH-500-1.5B / 3B	greedy diff	± 2.0	+6.4 / +5.8	gate **fail**: not comparable
S1	797fa96		**A-1 law on a non-Qwen model** (OLMo-2-1B, ARC-Challenge)	pooled r, $\sigma \leq 0.002$, 1600 pairs	≥ 0.5 GO; < 0.3 NO-GO	**0.790** [0.753, 0.824]	**GO**
S1	797fa96		B-1 winner per-item first-order (all parameters)	r over 600 fresh items	≥ 0.5 GO; < 0.3 NO-GO	0.162 [0.067, 0.252]	**NO-GO**
S1	797fa96		B-2 sign agreement on changed items	share, n = 52	≥ 0.70 supported; < 0.60 not	0.654	inconclusive
C	1437d44 6118222	+	validity; base gate	recomputed = printed; ± 2.0	exact; required	1018 = 1018; +1.5 / +0.5	valid; pass
C	1437d44 6118222	+	**C-1 same-run RandOpt (N = 5000, K = 50) − SC@50**	paired item bootstrap	CI > 0 RandOpt ahead; CI < 0 SC ahead	**−2.65 [−4.32, −0.99]**	**SC AHEAD**
C	1437d44 6118222	+	K = 10 (secondary)	same	same	+1.97 [0.00, 3.87]	no difference detected
C3B	e78cfd0		validity; base gate	recomputed = printed; ± 2.0	required	1143 = 1143; +0.9 / +0.5	valid; pass
C3B	e78cfd0		**C3B-1 same-run RandOpt − SC@50, GSM8K-3B**	paired item bootstrap	as C-1	**−1.59 [−2.65, −0.53]**	**SC AHEAD**
C3B	e78cfd0		K = 10	same	same	−1.06 [−2.35, 0.15]	no difference detected
G2	d2b3d68 5be1720	+	environment gate	base vs P2 greedy, same items	≤ 1.0 pp	0.16	valid
G2	d2b3d68 5be1720	+	**G2-1 same-run RandOpt − SC@50, GQA**	paired item bootstrap	as C-1	**+3.47 [+1.62, +5.41]**	**RANDOPT AHEAD**
G2	d2b3d68 5be1720	+	K = 10	same	same	+4.36 [+2.26, +6.54]	RandOpt ahead
G2R	9e84c4f 0754376	+	**G2R-1 GQA advantage with a disjoint population (seed 43)**	paired item bootstrap	REPLICATED if CI > 0	**+3.63 [+1.78, +5.49]**; $D_2 - D_1$ +0.16 [−0.97, +1.29]	**REPLICATED**
O1	e2e2400		**O1-1 same-run RandOpt − SC@50, GSM8K / OLMo-2-1B (non-Qwen)**	paired item bootstrap	as C-1	**+8.72 [+6.75, +10.77]**	**RANDOPT AHEAD** (Claim 1 model-dependent on GSM8K)
O1	e2e2400		K = 10; O1b second seed	same; budget rule	–	+7.58 [+5.31, +9.93]; O1b not run ($87 > $58.5)	RandOpt ahead; skipped by rule
PS	a1caf88		**PS: RandOpt K = 50 − SC@50 under the prompt chosen on RandOpt's selection set** (Qwen-1.5B / Qwen-3B / OLMo / GQA)	paired item bootstrap	as G4	**−2.96 [−4.62, −1.29] / −0.38 [−1.67, 0.91] / −23.65 [−26.23, −21.08] / −1.21 [−2.83, 0.40]**	**SC ahead / equivalent / SC ahead / n.d.: no row RandOpt ahead**
OS	c14ff18		**OS-1 OmniSpatial winner (direct) − BASE under the prompt chosen on SEARCH200 (manual_cot)**	image-cluster bootstrap (as M1)	as G4	**−1.50 [−5.35, +2.23]**	**NO DIFFERENCE DETECTED** (valid; answers identical to M1)
OS	c14ff18		OS-2 winner gain with the prompt held fixed (manual_cot)	same	reported	−0.67 [−3.95, +2.51]	no difference (under zeroshot_cot −3.83 [−6.81, −0.84])
GD	0bb89eb		**GD-1 RandOpt searched under the direct prompt − SC@50 direct, GQA (prompt held fixed)**	paired item bootstrap	as G4	**−0.32 [−1.29, +0.65]**	**NO DIFFERENCE, EQUIVALENT (± 2 pp)** (valid; finished on a second pod, cross-pod answers identical)
GB	77332dd		**GB-1 RandOpt searched under the boxed prompt − SC@50 boxed, GSM8K OLMo-2-1B (prompt held fixed)**	paired item bootstrap	as G4	**−2.12 [−3.49, −0.83]**	**SC AHEAD** (env and fidelity gates passed)
GB	77332dd		GB-2 same, Qwen2.5-1.5B	environment gate	± 1.0 pp	base selection 68.00 vs 73.00 printed by randopt.py	**INVALID-ENV, not run**
Q2	2a55ccc		**Q2-1/2 RandOpt − SC@50 with the plain prompt, Qwen 1.5B / 3B**	paired item bootstrap	as G4	**+2.58 [0.53, 4.62] / +7.05 [5.23, 8.95]**	**RANDOPT AHEAD** (plain prompt is worse for Qwen)
Q2	2a55ccc		damage prediction (all four rows)	plain − RandOpt-prompt base	supported if Qwen both < 5 pp	+4.70, −7.20 (GQA +11.3, OLMo +31.6)	**supported** (4/4 rows, locked measure); not decisive: vs the PS-chosen prompt Qwen-1.5B damage is +10.2 and RandOpt still lost
O2	6ecafcf		**O2-1 RandOpt (O1) − SC@50 with the plain question prompt, OLMo-2-1B**	paired item bootstrap	as G4	**−22.21 [−24.87, −19.56]**	**PROMPT-SC AHEAD** (valid env)
O2	6ecafcf		member fidelity gate; search on top of prompt	answer agreement ≥ 0.90, acc ≤ 2 pp	as locked	0.70 / 0.63 pp; −0.91 [−2.27, 0.53]	**gate failed** → members secondary not valid
G4	2ad4e46		**G4-1 RandOpt (CoT) − SC@50 with a direct-answer prompt, GQA**	paired item bootstrap	as C-1, + equivalence ± 2	**−1.21 [−2.83, +0.40]**	**NO DIFFERENCE DETECTED** (not equivalent)
G4	2ad4e46		direct-prompt BASE − CoT BASE; search on top of prompt	paired bootstrap	reported	+11.31 [8.48, 14.14]; −0.08 [−0.97, +0.81]	prompt effect; search adds nothing
G3	dc19fd4		**G3-1 does a 1024-token budget remove the GQA advantage?**	Δ = D256 − D1024; D1024	supported if Δ CI $>$ 0 and D1024 CI $\ni 0$	Δ +0.97 [0.32, 1.70]; D1024 +2.67 [0.73, 4.52]	**PARTIAL**

Study	Lock	Test	Statistic	Rule	Result	Outcome
G3	dc19fd4	G3-2 base vs members non-termination at 256	difference, CI	supported if CI > 0	+1.90 [0.31, 3.48]	supported (small)
S1-7B	d64a1ad	**7B-1 law at 7B** (OLMo-2-7B, ARC)	pooled r, $\sigma \leq 0.002$	≥ 0.5 GO; < 0.3 NO-GO	**0.787** [0.758, 0.820]	**GO**
RV	c33fec2	**RV-1 RandOpt − SC@50 under the public template chosen on the selection set** (Qwen-1.5B / Qwen-3B / OLMo / GQA)	paired item bootstrap	as PS	**+4.17 [+1.97, +6.44] / −0.45 [−1.67, +0.76] / −5.23 [−7.66, −2.81] / −2.67 [−4.60, −0.73]**	**RandOpt ahead / equivalent / SC ahead / SC ahead** (pooled six-candidate rule: SC ahead / equiv. / SC ahead / SC ahead)
RV	c33fec2	**RV-2a RandOpt searched under boxed − SC@50 boxed, Qwen2.5-3B (prompt held fixed)**	paired item bootstrap	as GB	**−1.36 [−2.35, −0.38]**	**SC AHEAD** (gates passed; resumed across sessions, amendment 1)
RV	c33fec2	**RV-2b same, Qwen2.5-1.5B** (amendment 2; GB-2 re-run with the lock's gates)	paired item bootstrap	as GB	**−4.32 [−5.91, −2.81]**	**SC AHEAD** (gates passed)
RV	c33fec2	**RV-3 second OLMo-2-1B search (seed 43), RandOpt − SC@50 under RandOpt's prompt** (amendment 3)	paired item bootstrap	as O1	**+10.39 [+8.26, +12.51]**	**RANDOPT AHEAD** (O1 replicated; gates passed)
RV	c33fec2	RV-4 randopt.py base print, Qwen2.5-1.5B, fresh pod	reproduction	reading fixed in the lock	randopt.py 73.00 (4 and 1 engines) vs our runner 68.00	**engine-specific base print**: fidelity limitation; exploratory R8b.1/R8b.7 recomputed
RV	c33fec2	RV-5 SC@50 at T = 0.5 / 1.0 (Qwen-3B boxed; GQA direct)	paired item bootstrap	reported; T = 0.7 stays primary	−0.23 / −1.14; −1.05 / +0.24 (all CIs $\ni 0$)	no outcome changes

B. Methods Details

Perturbations. RandOpt generates one Gaussian stream per seed and assigns each parameter tensor a prefix of it; fold in Equation 1 maps per-tensor gradients onto that shared stream. Every perturbation used in Sections 4–5 was reconstructed from RandOpt's generator and checked byte-exactly against the serving engine's weights (packed-tensor hashes). For the same-run rows, our GQA re-implementation and the fast GSM8K runner use RandOpt's `WorkerExtension` to apply and reset perturbations; the only stated deviation from RandOpt's script is an exact reset from a stored base copy instead of subtract-to-restore. Fidelity gates compare the rewards of the first 24 perturbations with those in RandOpt's own logs (mean absolute difference 0.0135–0.0181, no systematic offset).

Models and data. Qwen2.5-1.5B-Instruct, Qwen2.5-3B-Instruct, Qwen2.5-VL-3B-Instruct, OLMo-2-0425-1B-Instruct, OLMo-2-1124-7B-Instruct, Qwen2.5-VL-7B-Instruct and Qwen3-VL-8B-Instruct at pinned revisions; vLLM 0.11 in bfloat16; greedy decoding (temperature 0, maximum 1024 tokens for GSM8K and 256 for GQA, as RandOpt) for selection and for the selected models on test; self-consistency at temperature 0.7, top-p 1, 50 samples with a fixed seed. GSM8K questions and answers were checked identical across RandOpt's data files, the public dataset and every run (1319/1319). The GQA selection set is RandOpt's 200 selection questions; the test set is a hash-selected sample of testdev with every question whose image appears in the selection set removed (1238 questions).

Decision rules. Each lock states the primary comparison, the bootstrap, and the outcomes RANDOPT AHEAD, SC AHEAD or NO DIFFERENCE DETECTED (plus EQUIVALENT when the interval lies within ± 2 points), with gates on base reproduction (within 1–2 points of an earlier independent measurement), environment and fidelity, and a budget rule computed from a smoke run that stops a search rather than shrinks N.

C. Additional Same-Run Results

K=10**.** Under RandOpt's prompts: Qwen-1.5B $+1.97$ [0.00, $+3.87$] (no difference), Qwen-3B -1.06 [-2.35, $+0.15$] (no difference), GQA $+4.36$ [$+2.26$, $+6.54$] and $+4.68$ [$+2.50$, $+6.95$] (second population), OLMo $+7.58$ [$+5.31$, $+9.93$] and $+11.68$ [$+9.25$, $+14.10$] (second population). Against the prompt chosen on the selection set: -0.38 [-2.20, $+1.52$], -0.53 [-1.97, $+0.91$], -24.56 [-27.37, -21.83], -0.57 [-2.18, $+1.05$]; RandOpt is ahead in no row.

Vote curves ($K = 1/5/10/20/50$). Qwen-1.5B: RandOpt 68.2/74.6/77.2/77.5/77.2, SC 57.5/69.9/75.2/78.7/79.8. OLMo (RandOpt's prompt): RandOpt 40.2/49.0/50.1/53.0/52.5.

Robustness (exploratory). With strict answer scoring (boxed-only or ####-only on GSM8K, exact match on GQA) prompt-selected SC versus RandOpt on its own lenient scorer gives -7.81 [-9.63, -6.06], -4.78 [-6.07, -3.49], -26.00 [-28.58, -23.43] and -1.05 [-2.67, $+0.57$]; the lenient scorer under-credits boxed answers (Qwen-1.5B base 70.20 lenient vs 74.45 strict) and rescues RandOpt's prompt (60.05 vs 7.96 strict). McNemar tests with Holm correction give

$p = 0.0025$, 0.65, $\approx 4 \times 10^{-67}$ and 0.34 for the four rows; the GQA image-cluster bootstrap gives $[-2.79, +0.41]$.

Temperature (RV-5, pre-registered). SC@50 at $T = 0.5/0.7/1.0$ under the chosen prompt: Qwen-3B boxed 86.88 / 87.04 / 87.79, $D = -0.23$ $[-1.52, +1.06]$ / -0.38 / -1.14 $[-2.50, +0.23]$; GQA direct 64.54 / 64.70 / 63.25, $D = -1.05$ $[-2.67, +0.65]$ / -1.21 / $+0.24$ $[-1.53, +2.02]$. No outcome changes.

Stability of prompt selection (exploratory, no new generations). Bootstrapping the 200 selection questions (10,000 draws), the rule picks the same prompt in 99.6% (Qwen-1.5B), 95.5% (Qwen-3B), 77.5% (OLMo) and 99.6% (GQA) of draws; with 50 questions drawn without replacement, 89%, 73%, 61% and 94%; with 25, 70%, 52%, 52% and 82%. D = RandOpt − SC@50 under each candidate: Qwen-1.5B -2.65 / $+2.58$ / -2.96 (RandOpt's / plain / boxed), Qwen-3B -1.59 / $+7.05$ / -0.38, OLMo $+8.72$ / -22.21 / -23.65, GQA $+3.63$ / -1.21 / -0.57 (step-by-step / direct / short; step-by-step from the 256-token rerun). Weighting these by the bootstrap choice frequencies gives an expected D of -2.93, -0.19, -23.33 and -1.21 at 200 questions, and -2.53, $+0.53$, -23.09 and -1.08 at 50. Selection is stable where it matters at RandOpt's setting; smaller selection sets put Qwen-3B at risk of choosing the plain prompt, which is worse for Qwen models (Section 3.3).

Termination (G3). Raising the GQA token budget from 256 to 1024 shrinks RandOpt's advantage by 0.97 [0.32, 1.70] but leaves $+2.67$ $[+0.73, +4.52]$; at 256 tokens the base model fails to terminate more often than the selected models by 1.90 [0.31, 3.48] points. Termination explains about a quarter of the advantage.

RandOpt's printed base accuracy (RV-4). On a fresh pod, RandOpt's script prints a base selection reward of 73.00% for Qwen2.5-1.5B with 4 engines and with 1 engine; our engine measures 68.00% (and 68.5% in an independent earlier run) on the same 200 byte-identical prompts. The two engines agree on the perturbed rewards of the same 24 perturbations (signed mean difference -0.001) and on the test-set base (60.27 vs 60.05). For Qwen2.5-3B the print is 85.50 against our 84.00; for OLMo both give 41.50. We treat the print as an engine-specific effect of greedy bfloat16 decoding on this model, use the same-engine base for the exploratory selection-set statistics of Section 3.2, and note that a gate against this print caused our first Qwen-1.5B prompt-held-fixed search to be declared invalid (ledger: GB-2); the search was re-run under gates that compare like with like (RV-2b).

O2 member fidelity. Rebuilding O1's selected models with exact weight reset reproduced their answers on only 70.2% of items (accuracy within 0.63 points), most likely bfloat16 drift in RandOpt's subtract-to-restore; the "search on top of the prompt" secondary of that run is therefore not used, and Section 3.5 rests on searches run from scratch under the chosen prompt.

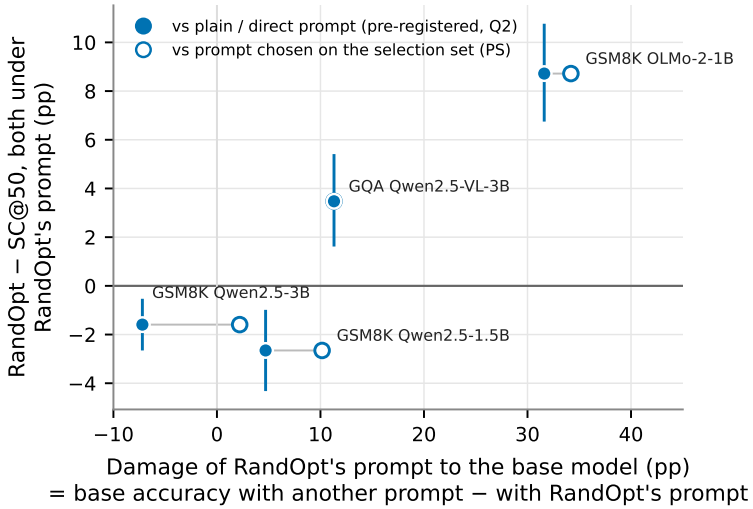


Figure 6. **Damage accompanies RandOpt's wins but does not decide them.** RandOpt minus SC@50 under RandOpt's prompt against the damage its prompt does to the base model, measured against a plain or direct prompt (filled, pre-registered) and against the prompt chosen on the selection set (open, post hoc). Qwen-1.5B is damaged by 10.2 points on the post-hoc measure, and RandOpt still lost.

D. Theory Derivations

Setup. $\theta' = \theta_0 + \sigma\epsilon$ with ϵ RandOpt's shared stream; for question j the answer margin is $c_j(\theta) = \text{logit}(\text{gold}) - \text{logit}(\text{strongest wrong answer at base})$. The first-order law (Equation 1) gives $\Delta c_j \approx \sigma\langle g_j, \epsilon\rangle$.

T1 (flip probability). Since ϵ is isotropic Gaussian, $\langle g_j, \epsilon\rangle \sim \mathcal{N}(0, \|g_j\|^2)$ over perturbations, so a random perturbation flips question j's two-way answer with probability $p_j(\sigma) = \Phi(-|c_j|/(\sigma\|g_j\|))$. Check (P0, Qwen2.5-VL-3B on GQA, 96×16 at $\sigma \le 0.002$, $\|g_j\|$ estimated from predictions only): AUC 0.935 for the 46 observed flips; 65 predicted versus 46 observed (ratio 1.41, within the fixed range [0.67, 1.5]).

T2 (unselected vote). Δc_j is symmetric about zero to first order, so each perturbation keeps the base answer with probability $\Phi(|c_j|/(\sigma\|g_j\|)) > 1/2$ and the majority of K unselected perturbations converges to the base answer. Check: the majority over 16 perturbations equals the base answer on 96/96 questions. Beyond first order, curvature adds a mean drift $\mathbb{E}[\Delta c_j] \approx \frac{1}{2}\sigma^2\,\text{tr}(\text{fold-Hessian})$; at $\sigma = 0.005$ the mean margin drift is -0.62 [-1.00, -0.27], T1 loses discrimination (AUC 0.61) and the vote departs from base on 2 of 96 questions, the same σ at which the law breaks.

T3 (selection as a first-order step). Let the selection score be, to first order, $S(\epsilon) \approx S_0 + \sigma\langle G_S, \epsilon\rangle$ with G_S the folded gradient of a smoothed selection accuracy. Conditioning a Gaussian ϵ on a high projection onto $u = G_S/\|G_S\|$ shifts only that component, $\mathbb{E}[\epsilon \mid \text{selected}] = \mu_{K/N}\,u$ with $\mu_{K/N}$ the mean of the top-K/N tail of $\mathcal{N}(0,1)$ (≈ 2.67 for the top 1%), and leaves the orthogonal components as fresh noise. For any question j a selected perturbation then gives $\Delta c_j \approx \sigma\mu_{K/N}\|g_j\|\cos(g_j, G_S) + \mathcal{N}\big(0, \sigma^2\|g_j\|^2(1 - \cos^2(g_j, G_S))\big)$. Gains transfer in proportion to a question's gradient alignment with the selection set, so shifts shared by many items (a format or an answer-preference tilt) transfer and item-specific fits do not; the vote of selected models keeps the common shift and, by T2, votes away the orthogonal noise. T3 is derived, not tested directly; the noise-plus-gradient predictor correlates with re-ranking gains at $r = 0.40$ and with fresh-item gains at 0.61, working through the content tilt.

T4 (perturbations and sampling). Temperature sampling flips question j with probability $1/(1 + \exp(|c_j|/T))$; a perturbation with $\Phi(-|c_j|/(\sigma\|g_j\|))$. Both decrease in the margin and differ only through $\|g_j\|$, which varies moderately across questions (coefficient of variation 0.29), so $\sigma\|g_j\|$ acts as a per-question effective temperature and the two mechanisms rank questions alike (model-implied Spearman 0.94). The $\|g_j\|$ heterogeneity and the persistent common component of T3 are what distinguish perturbations from re-sampling.

Not covered. Chain-of-thought accuracy (multi-step generation compounds token-level shifts nonlinearly), vision weights, $\sigma \ge 0.005$, and the OmniSpatial winner's residual gain.

E. Tilt-Law Details

Test	Model	Contrast	σ	r(predicted, measured)
Stage 2	Qwen3-VL-8B	front	0.002	0.938
R1	Qwen3-VL-8B	left	0.002	0.915
R2b	Qwen2.5-VL-7B	front	0.002	0.925
I5 (per item, pooled)	Qwen3-VL-8B	front	0.002	0.771
P0	Qwen2.5-VL-3B (LM-only)	per-question answer contrast	0.001	0.977
P0	Qwen2.5-VL-3B (LM-only)	per-question answer contrast	0.002	0.915
P0	Qwen2.5-VL-3B (LM-only)	per-question answer contrast	0.005	0.306

Per noise scale, the per-question law gives $r = 0.977/0.915/0.306$ at $\sigma = 0.001/0.002/0.005$ for Qwen2.5-VL-3B on GQA, $0.925/0.752/0.21$ for OLMo-2-1B and $0.914/0.755/0.25$ for OLMo-2-7B on ARC-Challenge. Vision: with the vision encoder perturbed, the first-order standard deviation of the vision group's contribution is 7.1 nats against a measured 1.5, and adding the vision term lowers the full-perturbation fit from 0.84 to 0.53. For the OmniSpatial winner evaluated per item on 600 fresh matched items with all parameters perturbed, the prediction gives $r = 0.162$ [0.067, 0.252] (failing its locked test); descriptively, the vision part carries 93% of the prediction's variance with $r = -0.008$, and the language part alone gives $r = 0.621$.

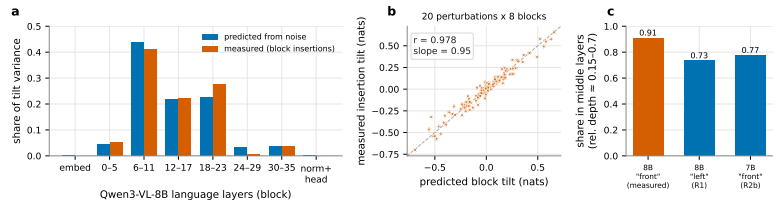


Figure 7. **The tilt lives in the middle language layers.** Share of the measured and predicted tilt variance by block for Qwen3-VL-8B (exact single-block insertions, 20 perturbations); the block-level prediction holds at $r = 0.978$ and the contributions add ($r = 0.96$).

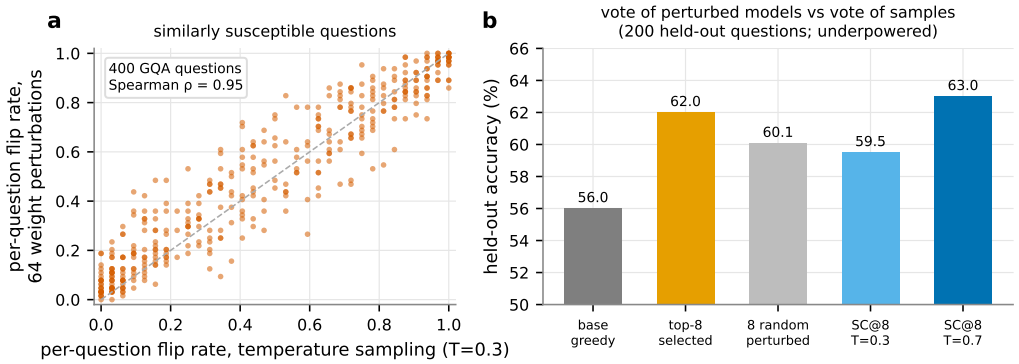


Figure 8. **The chain-of-thought regime.** (a) Per-question flip rates under 64 weight perturbations and under temperature sampling of the base model are similarly ranked (Spearman 0.946) on 400 GQA questions: similarly susceptible questions, not an equivalent mechanism (perturbed models carry persistent effects, $r = 0.74$). (b) On 200 held-out questions the vote of the top-8 selected perturbations (62.0) does not beat SC@8 (63.0 at $T = 0.7$; CI of the difference against SC at $T = 0.3$ $[-3.5, +8.5]$, underpowered).

F. OmniSpatial Case Study

A pre-committed N=5000 search on Qwen3-VL-8B (all parameters including vision; $\sigma \in \{0.00025, 0.0005, 0.001, 0.002\}$) on OmniSpatial perspective taking (Jia et al., 2026), direct-letter prompt, in three stages: a selection set of 200, a re-ranking set of 200 (543 candidates) and the official test set of 561 items. The winner (seed 9504111, $\sigma = 0.002$) gained $+2.5$ points on selection and $+8.0$ on re-ranking, and lost 2.67 on the official test. The selection and re-ranking sets are 185/200 and 189/200 eight-way compass-format items; the official test has 2 of 561 (Figure 9a). The re-ranking gain lies entirely on compass items. On fresh items drawn from the same source as the selection data (pre-registered, image-cluster bootstrap), the winner keeps $+2.67$ $[+0.17, +4.93]$; a gold-front concentration test is not supported ($+1.95$ $[-2.57, +6.47]$); the noise-plus-gradient predictor correlates with fresh gain at $r = 0.61$ $[0.42, 0.75]$; and the top 50 beat matched controls by $+1.16$ $[+0.31, +2.03]$. Removing the answer-content prior that selection favoured leaves $+2.33$ of the $+2.67$ (not explained). A prompt control (pre-registered) shows the gain to be specific to the direct prompt: under the benchmark's own step-by-step prompts the winner's gain is -0.67 $[-3.95, +2.51]$ and -3.83 $[-6.81, -0.84]$, and the unperturbed model under the prompt chosen on the winner's selection data reaches 38.83 against the winner's 37.33 (-1.50 $[-5.35, +2.23]$, no difference detected). Earlier conclusions drawn on the official test set before the split mismatch was found are superseded and kept in the repository as a record.

G. Cross-Paper Comparisons

Before the same-run rows, we compared SC@50 of the unperturbed model with the RandOpt numbers published by Gan & Isola (2026), gated on reproducing their base accuracy within 2 points. The gate passed for GSM8K with Qwen2.5-1.5B and 3B, where SC@50 scores $+3.4$ and $+1.1$ points above the published RandOpt and 5.7–10.7 points above the published test-time majority-vote baseline; it failed for GSM8K-0.5B, GQA and MATH-500, which are therefore not comparable. These comparisons are across papers and are superseded by Section 3.

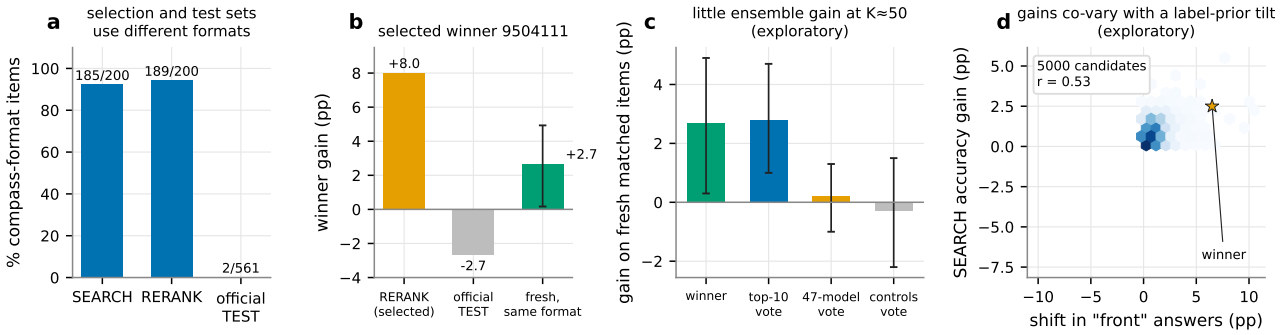


Figure 9. **The OmniSpatial winner.** (a) Selection and test sets use different formats. (b) The winner's gain by stage; on fresh same-format items $+2.67$ $[+0.17, +4.93]$. (c) Little measured ensemble gain at $K \approx 50$ (exploratory). (d) Across the 5000 candidates, selection gain co-varies with a tilt toward the answer content the selection labels reward ($r = 0.53$; exploratory).

Task	Model	Base (ours / paper)	Gate	SC@50	TT-MV†	RandOpt†	Verdict
GSM8K	Qwen2.5-1.5B	59.3 / 58.8	pass	79.8 [77.6, 82.0]	69.1	76.4	matches
GSM8K	Qwen2.5-3B	80.3 / 79.8	pass	88.2 [86.5, 90.1]	82.5	87.1	matches
GSM8K	Qwen2.5-0.5B	43.2 / 39.9	fail	56.0 [53.1, 58.6]	41.0	54.1	not comparable
MATH-500	Qwen2.5-1.5B	49.6 / 43.2	fail	62.4 [58.2, 66.6]	50.0	59.7	not comparable
MATH-500	Qwen2.5-3B	64.4 / 58.6	fail	73.8 [70.0, 77.6]	60.8	68.7	not comparable
GQA (2000)	Qwen2.5-VL-3B	54.0 / 56.6	fail	59.2 [57.1, 61.4]	–	69.0	not comparable

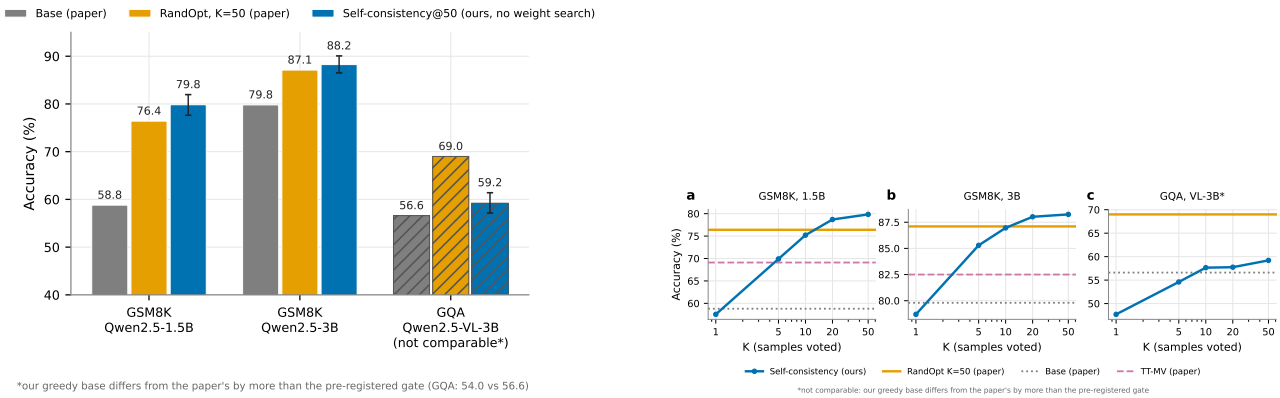


Figure 10. **Cross-paper comparison** (published RandOpt numbers, not same-run). Left: SC@50 versus the published RandOpt and majority-vote numbers on the gated rows. Right: SC@K as a function of K.

H. Prompts and Scorers

All prompts are applied through each model's chat template as a single user message; RandOpt's scorers extract the answer (GSM8K: the number after ####, else the last number in the response; GQA: the `boxed` content, else a short final answer or first line) and compare it with the reference. The candidates of Section 3.4, verbatim ({q} is the question):

- GSM8K, RandOpt's prompt: `{q} Let's think step by step and output the final answer after "####".`
- GSM8K, plain: {q}
- GSM8K, boxed: {q}\n`Please reason step by step, and put your final answer within \boxed{}.`
- GQA, RandOpt's prompt (step by step): `Look at the image and answer the question.` [two line breaks] `Question:` {q} [two line breaks] `Please reason step by step, and put your final answer within \boxed{}.`

- GQA, direct: the same with `Answer directly without explanation, and put your final answer within \boxed{}.`
- GQA, short: the same with `Answer the question using a single word or phrase, and put your final answer within \boxed{}.`

The public templates (fixed in the RV lock before any was run): lm-evaluation-harness `gsm8k_cot_zeroshot` (`Q: {q}\nA: Let's think step by step.`), lm-evaluation-harness `gsm8k` (`Question: {q}\nAnswer:`), the openai/simple-evals math template (`Solve the following math problem step by step. The last line of your response should be of the form Answer: $ANSWER (without quotes) where $ANSWER is the answer to the problem.\n\n{q}\n\nRemember to put your answer on its own line after "Answer:".`), the LLaVA-1.5 GQA template (`{q}\nAnswer the question using a single word or phrase.`) and the BLIP-2 template (`Question: {q} Short answer:`). Their base accuracies on the 200 selection questions were 60.0 / 65.0 / 67.5 (Qwen-1.5B), 77.5 / 72.0 / 89.0 (Qwen-3B), 8.0 / 31.0 / 58.0 (OLMo) and 67.0 / 64.0 (GQA, LLaVA / BLIP-2).

I. Compute

All runs used rented H100 80GB pods (vLLM 0.11). Recorded costs per study: same-run RandOpt searches with RandOpt's script, $46 (Qwen-1.5B), $52 (Qwen-3B) and $47 (OLMo); GQA searches $23 and $31 (second population); the prompt-held-fixed searches $21 (GQA) and $19 (OLMo); the reviewer round $87 (public templates, two Qwen prompt-held-fixed searches, a second OLMo population, the engine diagnosis and the temperature check); the prompt controls and prompt selection $3–6 each; the first-order-law measurements $1–5 each; the OmniSpatial transfer and prompt controls $6 and $2. The studies reported here sum to roughly $400 of GPU time, excluding the original OmniSpatial search.